Automatic Rebar Detailing System

Engineering Design Report

Project Details

Project Name: Regression Slab
Designer Name: Engineer
Date: 2026-09-08
Remarks: IS 456 slab regression

INPUTS

Length: 5 m
Width: 3 m
Slab Thickness: 150 mm
Support/Wall Thickness: 150 mm
Clear Cover: 25 mm
Floor Finish Load: 1.5 kN/m²
Live Load: 3 kN/m²
Concrete Grade: M35
Steel Grade: Fe500
Support Type: Simply Supported
Unit Weight of Concrete: 25 kN/m³
Preferred Bar Diameter: 12 mm

LOAD CALCULATION

Slab thickness: 150 mm
Self weight: 3.75 kN/m²
Floor finish: 1.5 kN/m²
Dead load: 5.25 kN/m²
Live load: 3 kN/m²
Service load: 8.25 kN/m²
Ultimate load: 12.375 kN/m²

GEOMETRY

Clear span: 2.85 m
Effective short span: 3 m
Effective long span: 5 m
Aspect ratio: 1.667
Slab classification: Two Way Slab
Effective span: 5 m
Effective span method: IS 456:2000 Clause 22.2: effective span = clear span + support width; for the provided span input, clear span = total span - support width and effective span = total span. Short and long directions calculated independently.

TWO-WAY MOMENT DESIGN

Coefficient short: 0.0497
Coefficient long: 0.0337
Mu short: 5.535 kN·m/m
Mu long: 3.753 kN·m/m
Moment equation: Mx = alphaX x w x lx^2; My = alphaY x w x lx^2 (IS 456 Table 26 coefficient convention).
Short span used for both moments, lx: 3 m
Coefficient source: IS 456:2000 Table 26 with linear interpolation between m = 1.6 and 1.7.

FLEXURAL DESIGN

SHORT DIRECTION
b: 1000 mm
d: 121 mm
Mu: 5.535 kN·m/m

Ast required: 180 mm²
Ast minimum: 180 mm²
Ast provided: 182.78 mm²
Bar arrangement: 8 mm @ 275 mm c/c
xu: 6.31 mm
xu,max: 55.66 mm
Moment capacity: 9.41 kN·m/m
Status: PASS

LONG DIRECTION
b: 1000 mm
d: 113 mm
Mu: 3.753 kN·m/m
Ast required: 180 mm²
Ast minimum: 180 mm²
Ast provided: 182.78 mm²
Bar arrangement: 8 mm @ 275 mm c/c
xu: 6.31 mm
xu,max: 51.98 mm
Moment capacity: 8.77 kN·m/m
Status: PASS

SHEAR

Critical section: At d = 121 mm from the support face; one-metre strip reaction = 18.563 kN/m.
Vu: 17.065 kN/m
tau_v: 0.141 N/mm²
tau_c: 0.3008 N/mm²
tau_c,max: 3.7 N/mm²
Tension steel percentage for Table 19: 0.1511 %
Concrete shear capacity: 36.402 kN/m
Reinforcement: Not required / not applicable because concrete shear capacity is adequate.
Status: PASS

DEFLECTION

Basic L/d: 20
Tension steel percentage: 0.1511 %
Modification factor: NOT CALCULATED — IS 456 Fig. 4 chart data is not present in the configured design constants.
Actual L/d: 41.32
Permissible L/d: NOT CALCULATED — REQUIRED IS 456 modification-factor data is not implemented.
Status: FAIL
Method/status: Deflection check failed because the IS 456 Fig. 4 modification-factor method cannot be completed from the configured data.

DETAILING

Short reinforcement: 8 mm @ 275 mm c/c
Long reinforcement: 8 mm @ 275 mm c/c
Short spacing check: PASS
Long spacing check: PASS
Cover: 25 mm
Short maximum spacing: NOT CALCULATED — REQUIRED INPUT/METHOD NOT IMPLEMENTED mm
Long maximum spacing: NOT CALCULATED — REQUIRED INPUT/METHOD NOT IMPLEMENTED mm
Short clear spacing: NOT CALCULATED — REQUIRED INPUT/METHOD NOT IMPLEMENTED mm
Long clear spacing: NOT CALCULATED — REQUIRED INPUT/METHOD NOT IMPLEMENTED mm
Corner torsion reinforcement: Not required — corners are assumed free to lift; torsion reinforcement is required only where corner lifting is restrained.
Corner torsion status: Not required

FINAL CHECKS

Effective span validity: PASS | actual 5 | limit 0 | Effective span is finite and valid.

Slab classification: PASS | actual 1.667 | limit 2 | Aspect ratio = 1.667 -> Two Way Slab.
Short-direction flexural capacity: PASS | actual 9.41 | limit 5.535 | MuProvided 9.41 >= required 5.535 and xu <= xu,max.
Long-direction flexural capacity: PASS | actual 8.77 | limit 3.753 | MuProvided 8.77 >= required 3.753 and xu <= xu,max.
Short required reinforcement: PASS | actual 182.78 | limit 180 | AstProvided 182.78 >= AstRequired 180.
Long required reinforcement: PASS | actual 182.78 | limit 180 | AstProvided 182.78 >= AstRequired 180.
Short minimum reinforcement: PASS | actual 182.78 | limit 180 | AstProvided 182.78 >= AstMin 180.
Long minimum reinforcement: PASS | actual 182.78 | limit 180 | AstProvided 182.78 >= AstMin 180.
Short maximum spacing: PASS | actual 275 | limit 300 | Provided spacing 275 <= max 300.
Long maximum spacing: PASS | actual 275 | limit 300 | Provided spacing 275 <= max 300.
Shear check: PASS | actual 0.141 | limit 0.3008 | Shear stress is within concrete shear capacity.
Deflection control: FAIL | actual 41.32 | limit NOT CALCULATED — REQUIRED IS 456 modification-factor data is not implemented. | Deflection check failed because the IS 456 Fig. 4 modification-factor method cannot be completed from the configured data.
Cover and detailing: PASS | actual 25 | limit 150 | Nominal cover 25 mm is valid, slab thickness 150 mm.
Final status: NOT APPROVED